

Model similarity depends on the measurement.

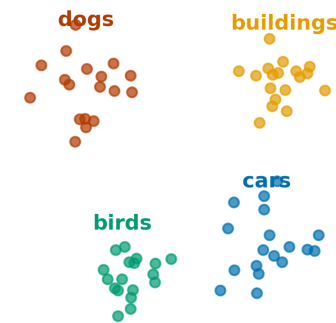
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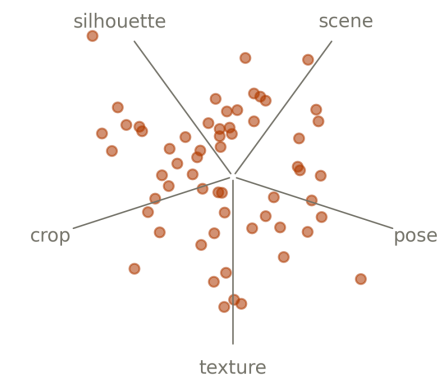
One probe of similarity: compare the geometry induced by a stimulus set.

Change the set and the answer changes. Broad categories give models an easy way to agree. Restrict to one superclass and the comparison shifts to processing strategy.

broad stimulus set



dog superclass only



same CKA formula, different stimulus question

CKA mixes category agreement with stimulus-level agreement.

$$X = M + W$$

M: between-class means W: within-class deviations

$$K_C = H(MM^T + MW^T + WM^T + WW^T)H$$

Many classes $\rightarrow$ HMM^TH dominates. All classifiers agree on it.

$k = 1 \rightarrow$ M vanishes after centering

$$K_C = HWW^TH$$

Now CKA measures only within-class structure, where architectures diverge.

Many categories

The HMM^TH term encodes how each model arranges class centroids. Every competent classifier agrees on this, so CKA stays high.

One superclass

M collapses to a single point. Centering maps it to zero. What remains is WW^T : pose, texture, background, breed. CNNs and ViTs organize these differently.

What the competition fixed, and what it kept hidden.

Both tracks use CKA on a hidden evaluator. Local work had to use proxy models and proxy images, so the real test was whether a measurement story transferred beyond the visible setup.

Fixed

The score is mean pairwise CKA over centered hidden embeddings.

Hidden

The evaluator controls the final models, layers, preprocessing, and resolutions.

Submitted

Only JSON: 20 model-layer pairs for Blue, 1,000 ImageNet image IDs for Red.

Blue Team task

Choose 20 models that look maximally aligned.

The object being optimized is not accuracy. It is the density of a 20-model subgraph in CKA space.

INPUT

141 vision models with fixed representation layers



CHOICE

20 model-layer pairs



SCORE

maximize mean pairwise CKA

Red Team task

Choose 1,000 images that make models disagree.

Here the models are fixed. The intervention is the stimulus set: find images where the same architectures induce different representational geometries.



Centering does not make the stimulus set neutral.

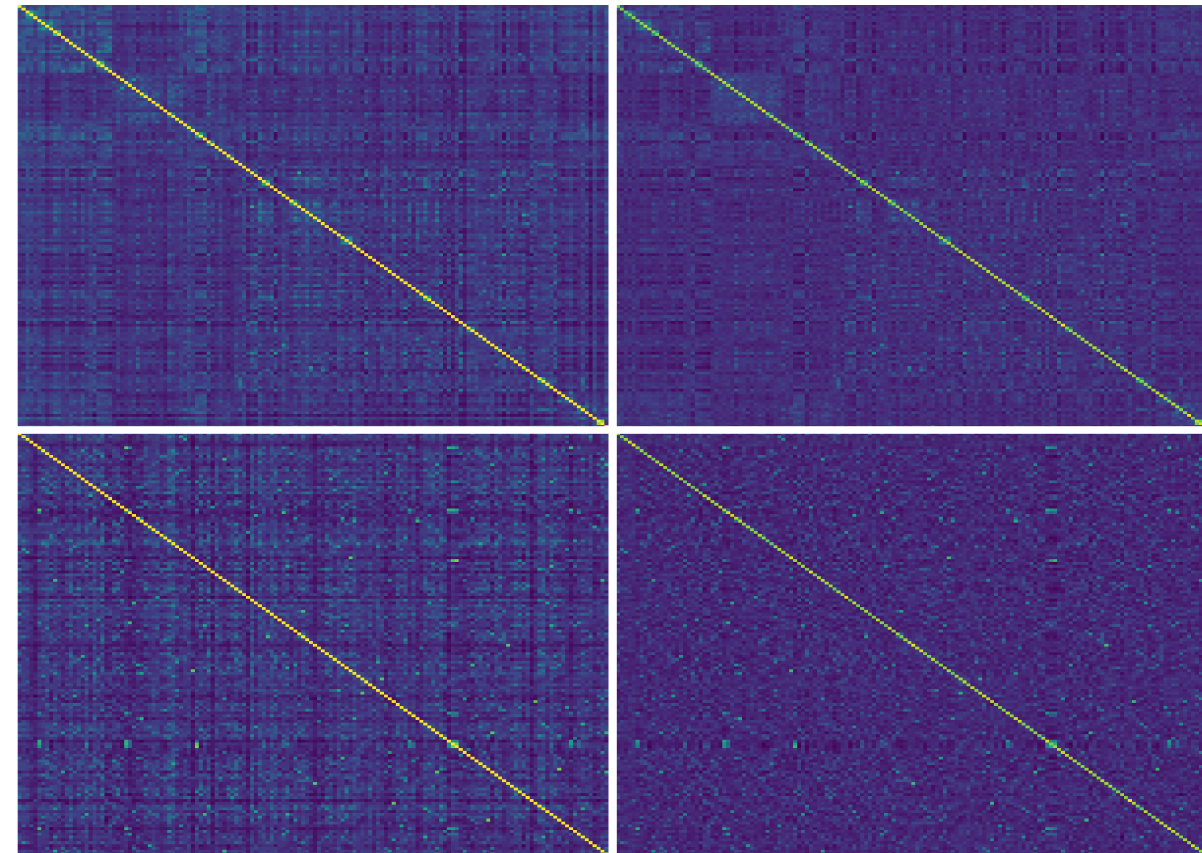
On multi-class images the block skeleton survives centering. On dog-breed images it vanishes, exposing the residual geometry models use inside a category.

Top row

Mixed ImageNet classes. Block structure visible before and after centering.

Bottom row

Dog breeds only. No blocks; residual within-class geometry is what remains.

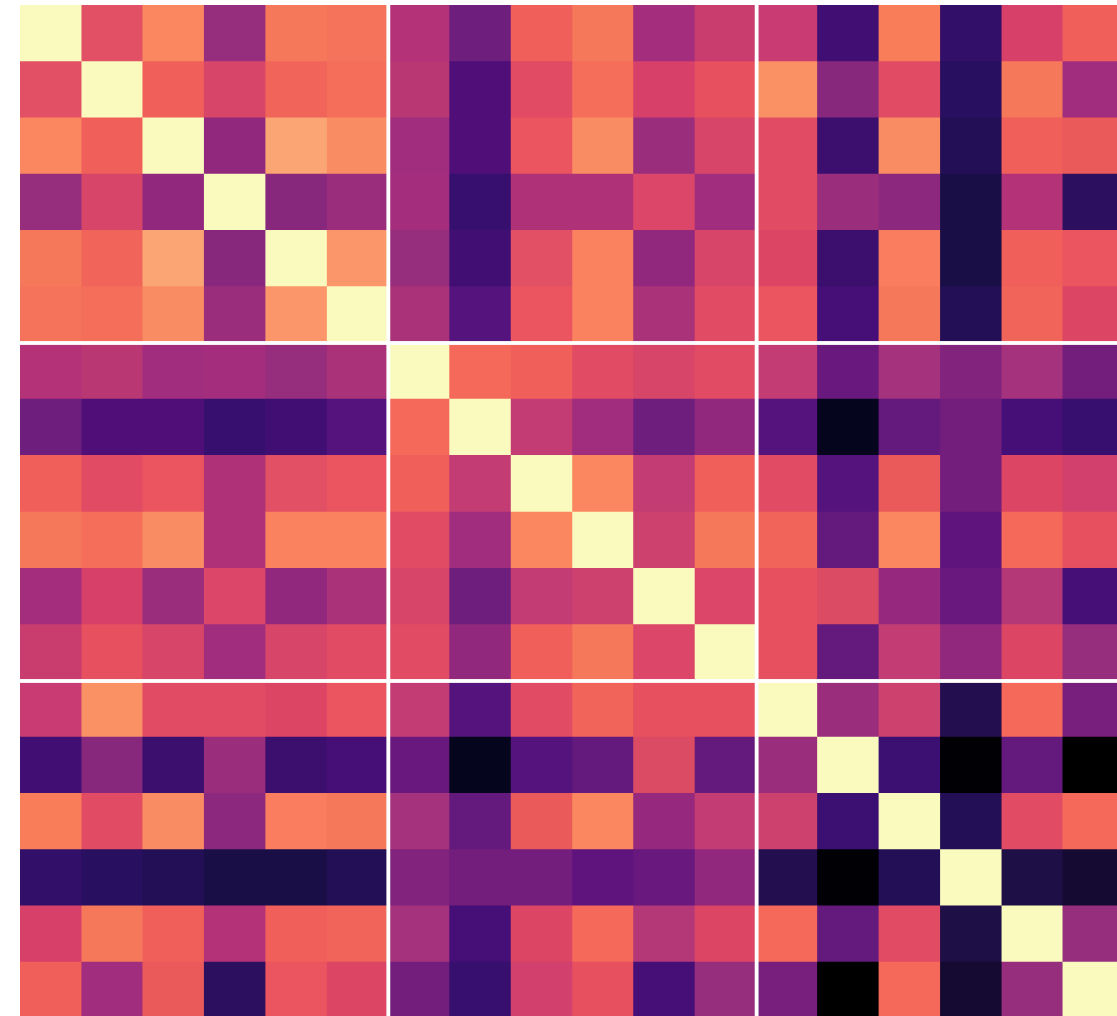


Real ResNet50 embeddings. Left column: K. Right column: HKH.

An architecture-level view: which models look similar under CKA?

Model selection becomes a densest-20-subgraph problem over the proxy CKA matrix. Under this view, similarity concentrates in ViT-family models: 18 of 20 are attention-based architectures.

The white grid lines in the figure separate architecture families. The bright top-left block is the high-CKA attention-family region. Cooler off-diagonal blocks show lower cross-family alignment.



Real pairwise CKA from actual model embeddings. Text removed; family blocks explained verbally.

A stimulus-level intervention: remove the easy contrast.

Dogs suppress broad category agreement. What remains is pose, texture, crop, background, and scene: axes that CNNs, ViTs, CLIP models, and masked autoencoders organize differently.

125

dog + canid synsets in ImageNet

6,250

candidate images

5,000

scored pool after divergence filter

1,000

submitted after 100k SA iterations

11 proxy models spanning CNN, ViT, CLIP, DINO, MAE, MLP-Mixer, and MambaOut.

Read CKA as a conditional measurement.

Red hidden score 0.547 vs baseline 0.479. The Platonic Representation Hypothesis says models are converging (Huh et al., 2024). This view suggests some convergence reflects shared category agreement. Restrict the stimuli and it weakens.

Red baseline

0.479

Red hidden eval

0.547

Red proxy

0.584

Blue submission

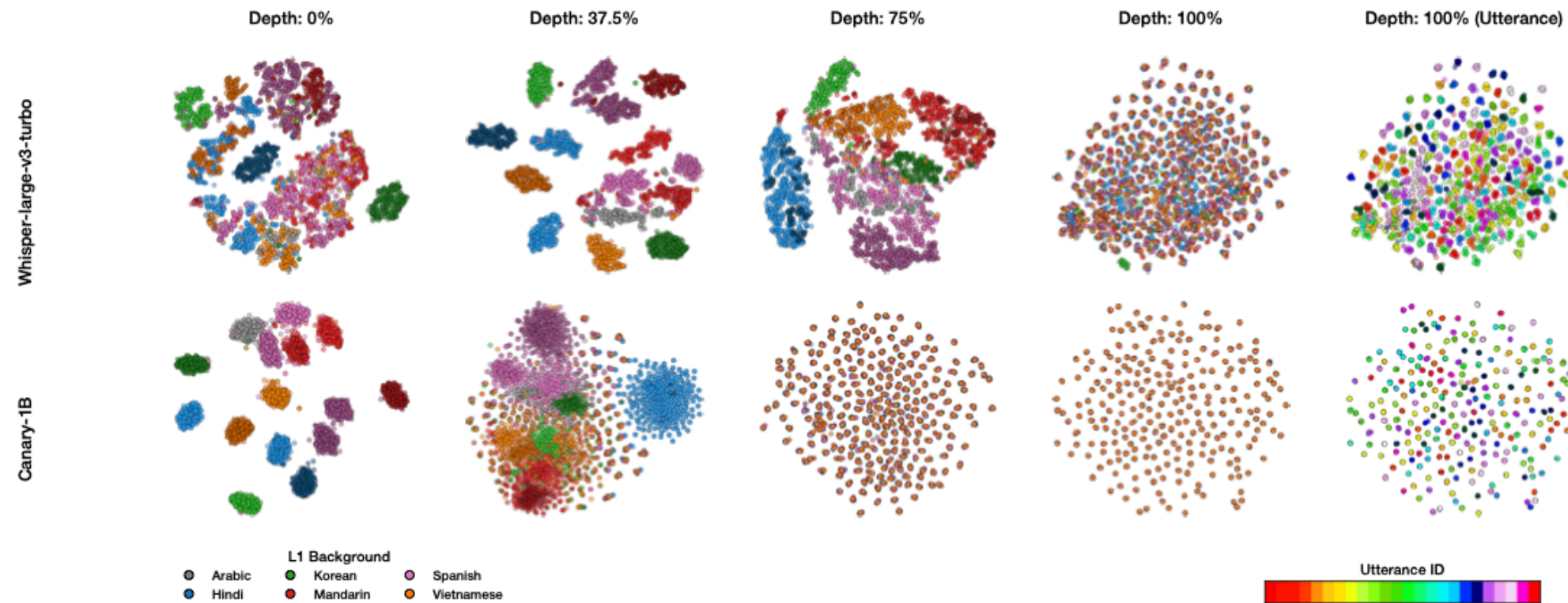
18/20

attention-family models

Score = 1 – mean pairwise CKA. Proxy overestimated hidden, but the direction transferred.

Processing strategy in speech: categorize early, integrate late.

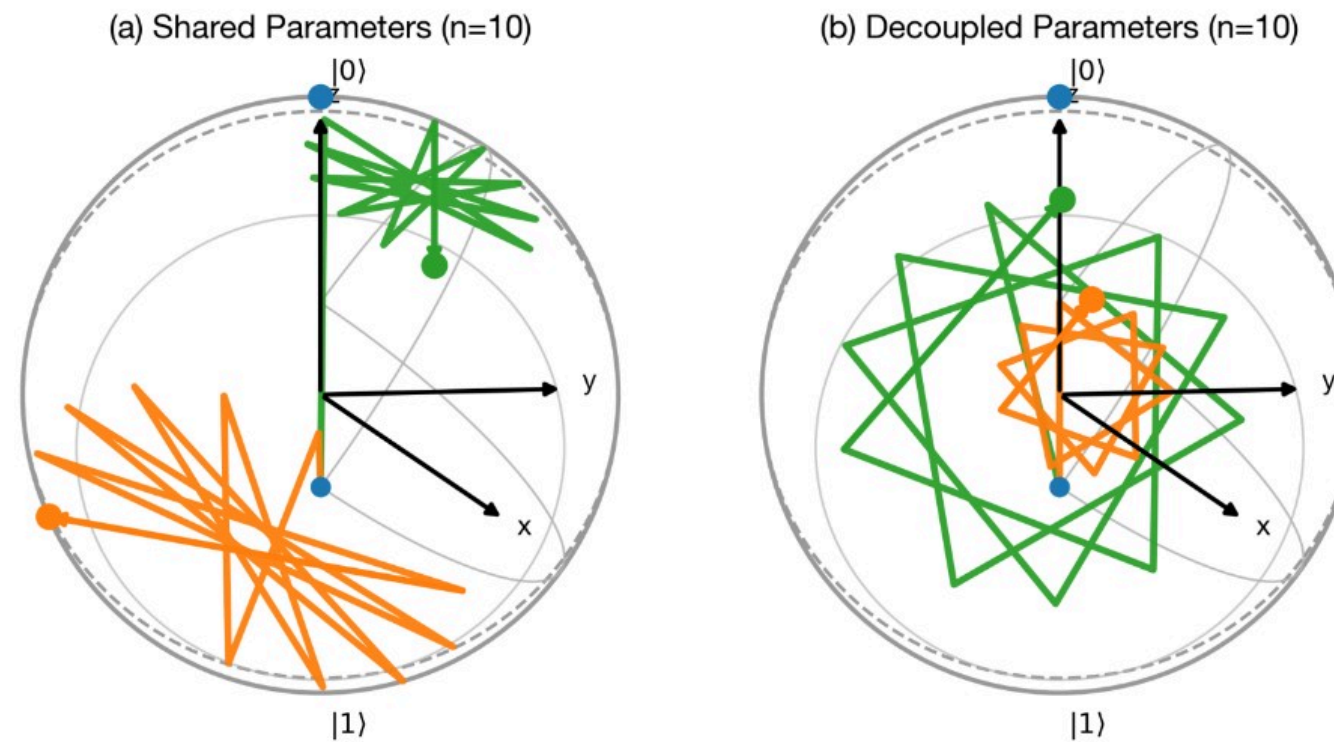
Layerwise probes ask where information becomes usable. The same utterances separate, merge, or stay mixed at different depths depending on the encoder.



Two ASR encoders, same utterances. Color reveals whether depth organizes the representation by background language or utterance identity.

Processing strategy in quantum language models: entanglement as memory.

A strategy probe asks whether the circuit uses a new computational route or embeds a classical one. Shared and decoupled parameterizations produce different state-space trajectories.



Bloch-sphere trajectories for shared versus decoupled parameters. The geometry makes the processing strategy visible.

A behavioral lens: compare failures, not activations.

TAB scores humans and LLMs in the same 19-symptom space, validated against expert SLPs. The question becomes: when the system is damaged, does it fail in human-like coordinates?

Common measurement space

Human aphasia transcripts and model outputs are scored with the same symptom battery.

Failure coordinates

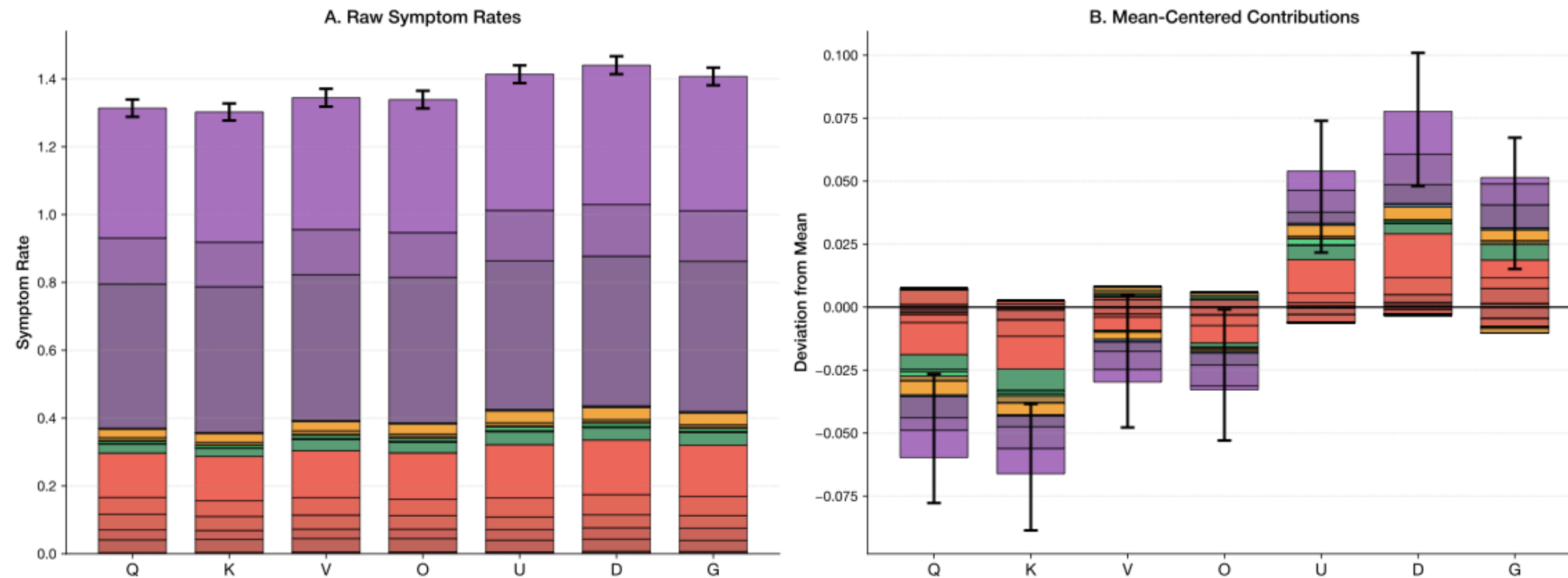
Similarity is not "same answer." It is whether damage moves behavior along the same axes.

Use case

Lesions turn architecture into evidence: attention and FFN damage should produce different symptom profiles if they support different computations.

A lesion lens: processing strategy revealed by damage.

112k lesioned outputs show that attention and FFN damage produce distinct, depth-stratified failure regimes. The symptoms are recognizable, but the mixture is machine-specific rather than a clean copy of classical aphasia.



Raw symptom rates and mean-centered contributions by component. Different lesions move the model through different symptom coordinates.

Similarity is not one thing.

A model can look aligned under one probe and diverge under another. The right question is not whether two systems are similar, but which processing strategy the measurement is sensitive to.

Stimuli

Broad categories inflate agreement. Fine-grained sets expose residual structure.

Depth

Speech models can solve the same task while making information usable at different layers.

State and damage

Quantum trajectories and lesion symptoms reveal strategies that output accuracy hides.